

PREPARATION SHEET LAB 01

SAMPLING AND QUANTIZATION

1/1	Team members	
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Please fill out the document electronically – including the yellow fields above – and upload it to Moodle once per team by the start of the lab event (no matter who uploads it, it will automatically be assigned to all members).

Please note that the complete lab assignment can be found in another document.

Prep task (for lab entry test)

Familiarize yourself with the concepts of the chapter “DP01: Digitization and Digital Signals”, particularly

- sampling, sampling frequency, aliasing and quantization,
- DSP system UniDAQ2 board, interrupt-based sample-by-sample processing in C
- rounding of fixed-point numbers and techniques in C to avoid overflows after arithmetic operations

These topics will be addressed by the lab entry test at the beginning of the lab session.

Prep task 1: Interrupt handler and bit manipulation

- Which decimal(!) values are output after bit manipulation to channel 0 and channel 1 of the DAC, if the hexadecimal values received from ADC in the format `int16_t` were 0x5455, 0x80CC, 0x1234, 0x5678, 0x90AB, 0xCDEF, 0x1011, 0x1213

- Channel 0: 0x5455 in hex, 21589 in decimal
- Channel 1: 0x80CC in hex, -32564 in decimal

Channel	ADC input	Mask	Result hex	Result decimal
0	0xFC7F	0x5555	0x5455	21589
1	0x83EE	0xC000	0x8000	-32564

Prep task 2: Sampling and quantization

- Determine the sampled discrete-time signal $x[n]$ (without quantization).
- Determine the eight signal values $x[n]$, $\hat{x}[n]$, $n = 0, \dots, +7$ before and after 4-bit

- sampled discrete-time signal: 1, 0.7071, 0, -0.7071, -1, -0.7071, 0, 0.7071
- sampled and quantized signal: 0.875, 0.625, 0, -0.750, -1, -0.750, 0, 0.625

n	$x[n]$ exact	$x[n]$ decimal	$\hat{x}[n]$ after 4-bit truncation
0	1	1.0000	0.875
1	$\frac{\sqrt{2}}{2}$	0.7071	0.625
2	0	0.0000	0.000
3	$-\frac{\sqrt{2}}{2}$	-0.7071	-0.750
4	-1	-1.0000	-1.000
5	$-\frac{\sqrt{2}}{2}$	-0.7071	-0.750
6	0	0.0000	0.000
7	$\frac{\sqrt{2}}{2}$	0.7071	0.625

Table 2: First eight samples before and after 4-bit quantization